

6201-090301b Типовый пример
 Задача на поиск экстремума

$$F = x_1 + x_2 + x_3$$

$$x_1 \cdot x_2 \cdot x_3 = 1$$

$$(x_1, x_2) \geq 0$$

$$\Phi(x) = F(x) + \sum_{i=1}^n \lambda_i (\phi_i(x) - b_i)$$

$$\Phi(x_1, x_2, x_3) = x_1 + x_2 + x_3 + \lambda (x_1 x_2 x_3 - 1)$$

$$(x_1, x_2) \geq 0$$

$$n=3 \quad m=1$$

$$\begin{cases} 1 + \lambda(x_2 \cdot x_3) = 0 \\ 1 + \lambda(x_1 \cdot x_3) = 0 \\ 1 + \lambda(x_2 \cdot x_1) = 0 \\ x_1 \cdot x_2 \cdot x_3 - 1 = 0 \\ (x_1, x_2) \geq 0 \end{cases}$$

$$x_1 = x_2 = x_3 = 1$$

$$\lambda = -1$$

$$H^0 = \begin{pmatrix} 0 & 0 \\ 0 & H \end{pmatrix}$$

$$H = \frac{\partial^2 \Phi}{\partial x_i \partial x_j}$$

$$Q = (H \quad 1 \quad 1)$$

$$Q' = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$Q = \begin{vmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \dots & \frac{\partial f_n}{\partial x_n} \end{vmatrix}$$

$$H = \begin{vmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & -1 & -1 \\ 1 & -1 & 0 & -1 \\ 1 & -1 & -1 & 0 \end{vmatrix} \quad D_1 = 0 \quad D_2 < 0 \quad D_3 < 0 \quad D_4 < 0$$

$$-1^{n+1}P = 1 - 1/2^2 > 0 = 7 \text{ is max}$$

$$f^m = -1^1 = -1$$

Answer: max $D_2 = D_3 = D_4 = 1 - 1/2^2 \Rightarrow$ max
 $(1, 1, 1)$ - min

1/2

A. 5. max. Minimization. The feasible region

$$x_1^2 + x_2^2 \leq 2.5$$

$$F = 3x_1 + 4x_2 \rightarrow \max$$

$$\begin{cases} x_1^2 + x_2^2 \leq 25 \\ x_1 \cdot x_2 \geq 4 \\ (x_1, x_2) \geq 0 \end{cases}$$

$$\begin{cases} 3x_1 + 4x_2 = F \\ x_1^2 + x_2^2 = 25 \end{cases} \quad \begin{cases} M = \frac{F}{3} - \frac{4x_2}{3} \\ \frac{F}{3} - \frac{4x_2}{3} + x_2^2 = 25 \end{cases}$$

$$F^2 - 100x_2 + 25x_2^2 = 225$$

$$D = 64F^2 - 100(F^2 - 225) > 0$$

$$D > 0 \quad F^2 > 625$$

$$F = \pm 25$$

$$F = -25 \text{ - is not}$$

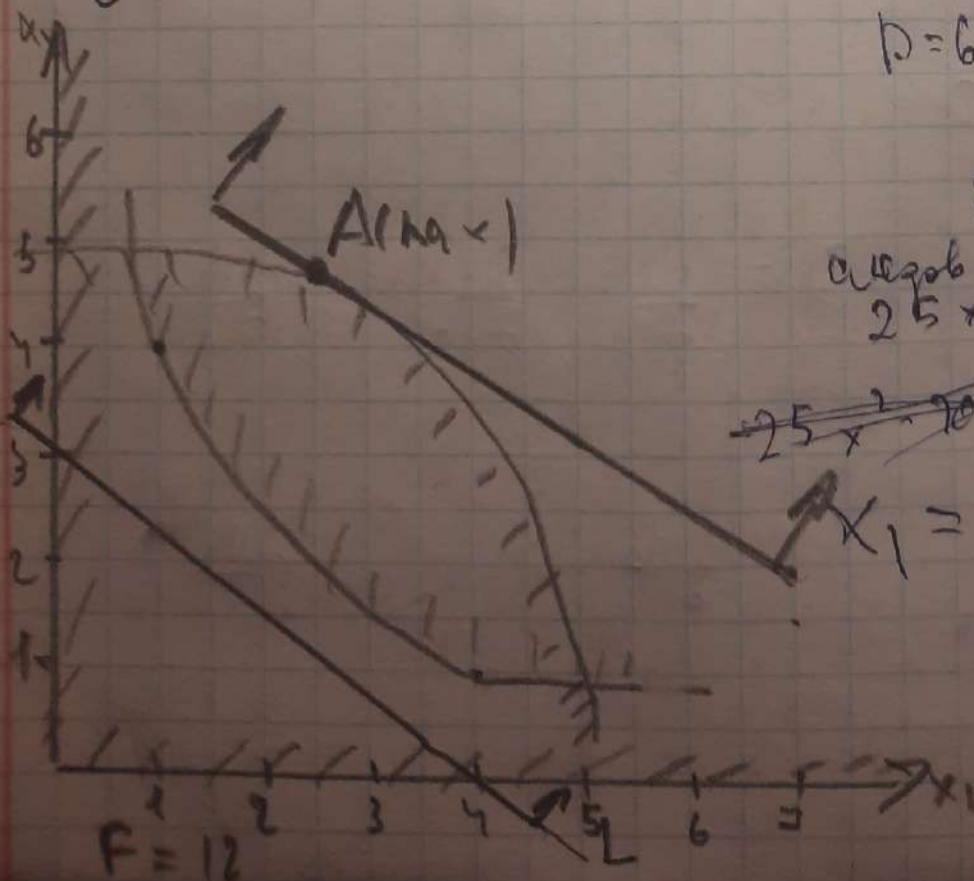
$$\text{answer } F = 25$$

$$25x_1^2 = 200x_1 + 400 = 0$$

$$x_2^2 - 8x_2 + 16 = 0$$

$$25x^2 - 100x$$

$$x_1 = 3 \quad x_2 = 4$$

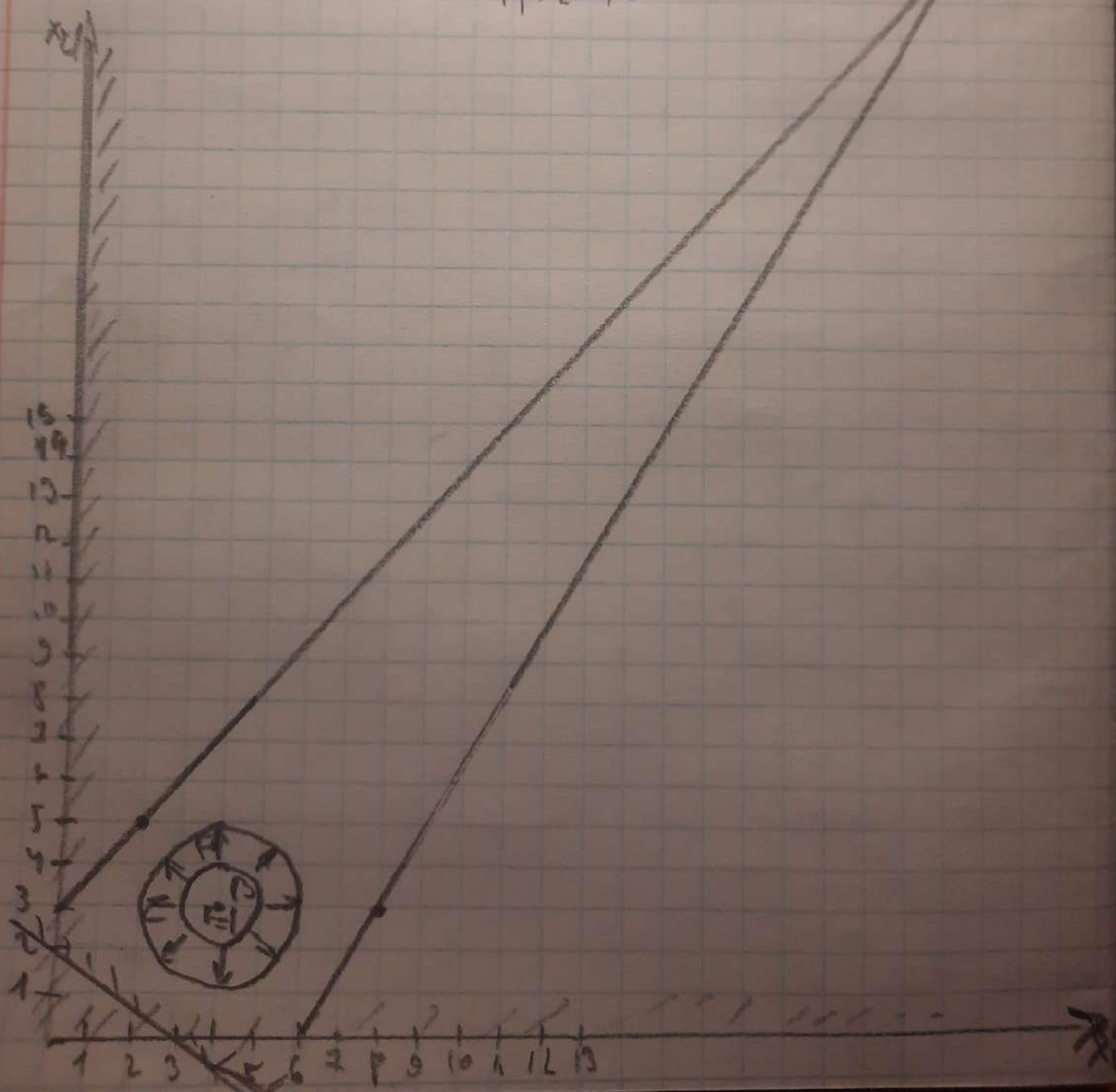


$$\frac{x_1}{4} + \frac{x_2}{3} = 1$$

Problem: move A - max. horizontal to (3,4) if $F=25$

$$F = (x_1 - 4)^2 + (x_2 - 3)^2 - \max(\min)$$

$$\begin{cases} 2x_1 + 3x_2 \geq 6 \\ 8x_1 - 2x_2 \leq 18 \\ -x_1 + 2x_2 \leq 8 \\ (x_1, x_2) \geq 0 \end{cases} \quad \begin{cases} \frac{x_1}{3} + \frac{x_2}{2} \geq 1 \\ \frac{x_1}{6} - \frac{x_2}{9} \leq 1 \\ -\frac{x_1}{8} + \frac{x_2}{4} \leq 8 \\ (x_1, x_2) \geq 0 \end{cases}$$



При $F=1$ $(3, 3)$, $(4, 4)$

при $F=4$ $(2, 3)$, $(4, 5)$ значения F будут

max в точке пересечения $(2) \cap (3)$ ограничена Т. А;

а min при $F=0$ в точке $(4, 3)$ Т. В

$$\begin{cases} x_1 = 2x_2 - 8 \\ 8x_2 - 24 - 2x_2 = 18 \end{cases}$$

$$4x_2 = 42$$

$$x_2 = 10,5$$

$$x_1 = 21 - 4 = 13$$

Ответ: Т. А - max $x_1 = 13$, $x_2 = 10,5$;

Т. В - min $x_1 = 4$, $x_2 = 3$